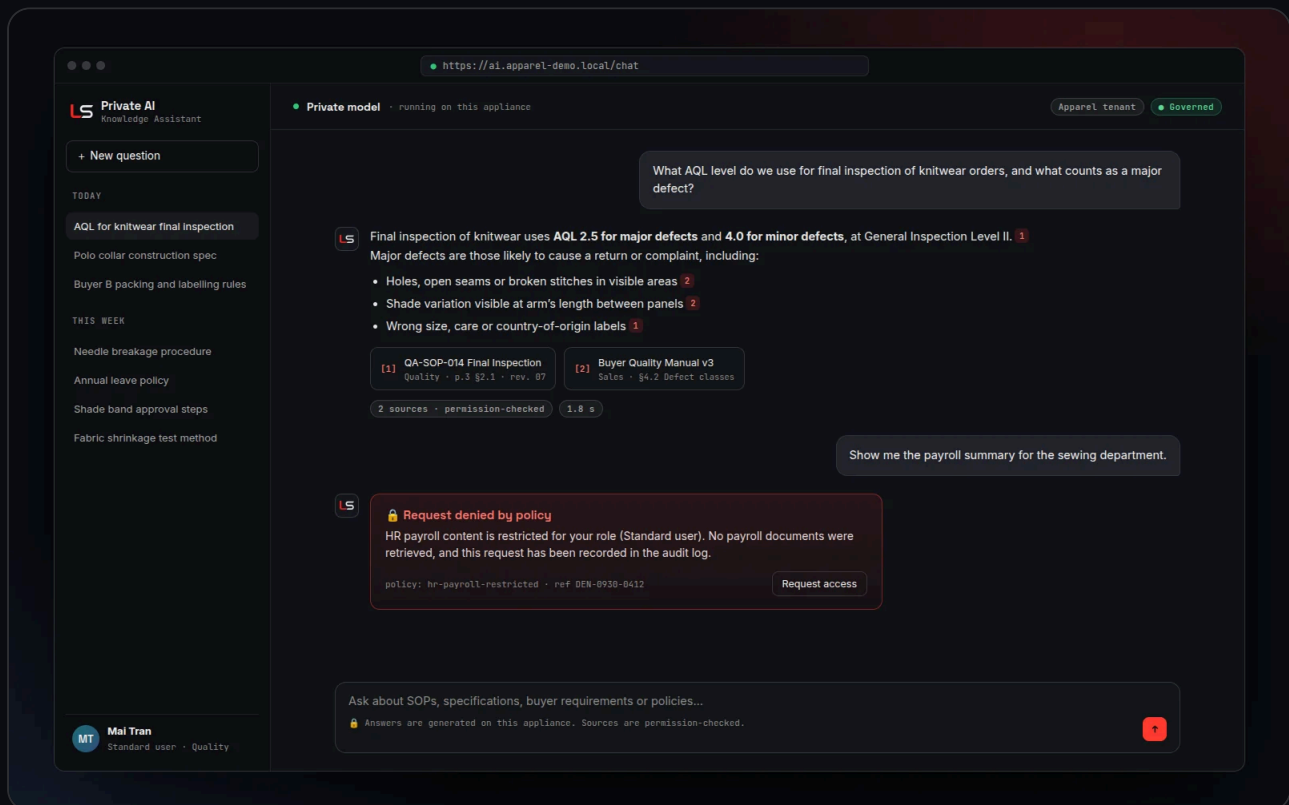


Private Knowledge Assistants

Cited, permissioned and audited answers from your own documents.



Answers from your own documents. Cited, permissioned and audited.

Every company runs on knowledge locked in documents: standard operating procedures, specifications, buyer manuals, policies and reports. Finding the right answer is slow, so employees increasingly paste company documents into public AI tools. That creates a data-control problem most boards have not yet addressed.

A private knowledge assistant solves both problems at once. Staff ask questions in plain language and receive answers drawn only from documents they are allowed to see, each with a citation to the source. The model, the index, the policies and the audit trail all run on hardware the company controls.

Key takeaways

- ✓ Faster, verifiable answers: every response cites the exact source passage.
- ✓ Access enforced before retrieval: restricted content fails closed and every attempt is logged.
- ✓ One console for governance: policies, access requests, activity and audit in a Security Center.

This paper explains how the Private Knowledge Assistant works, the controls that keep data private, where it delivers value first, and how to evaluate and roll it out.

■ THE CHALLENGE

Your people already ask public chatbots about company documents.

Staff paste SOPs, buyer requirements and costing sheets into public AI tools because finding the right answer internally takes too long. A private knowledge assistant gives them faster answers than a public tool, from sources they can check, without the data ever leaving the business.

- ✓ Knowledge is scattered across shared drives, email and legacy systems, so answers depend on who you know.
- ✓ Public chatbots are convenient but process prompts and uploads outside company control.
- ✓ Generic enterprise search ignores permissions nuance and returns documents, not answers.
- ✓ Without an audit trail, nobody can show auditors or customers how AI is being used.

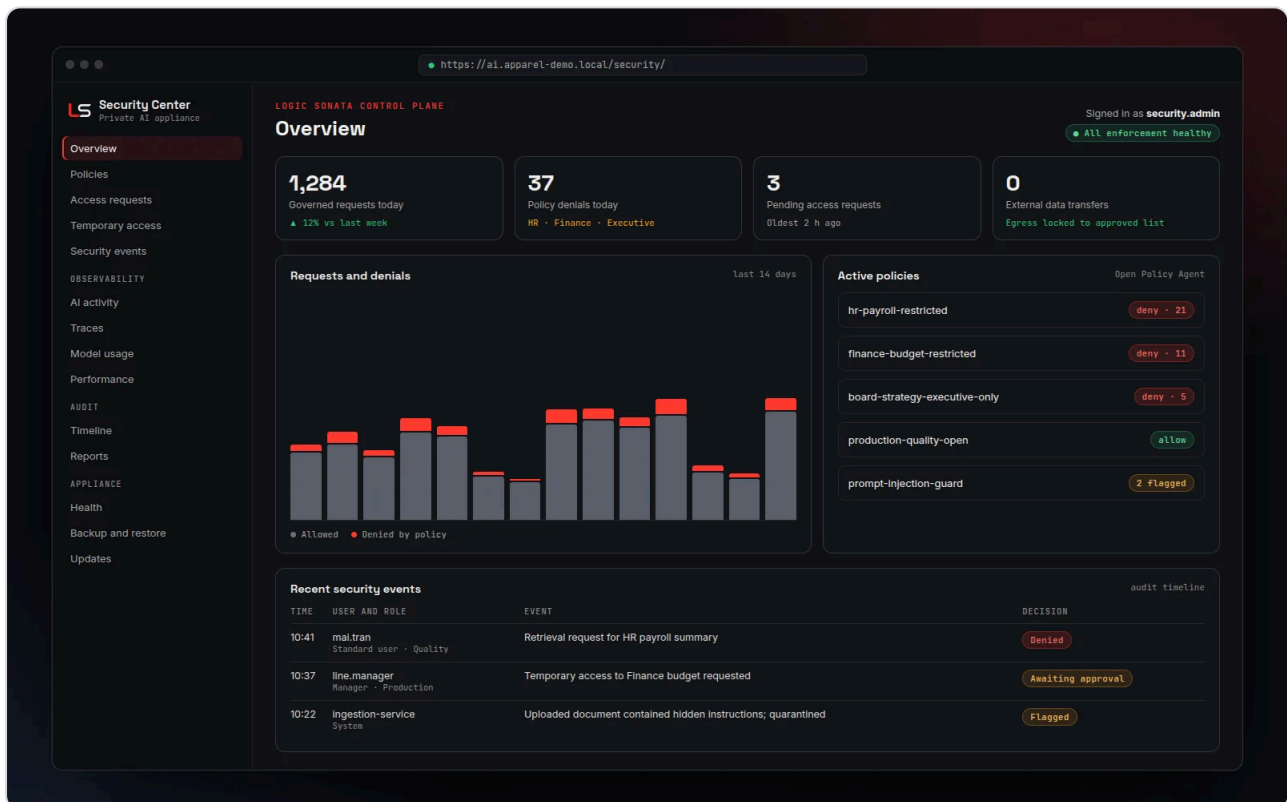
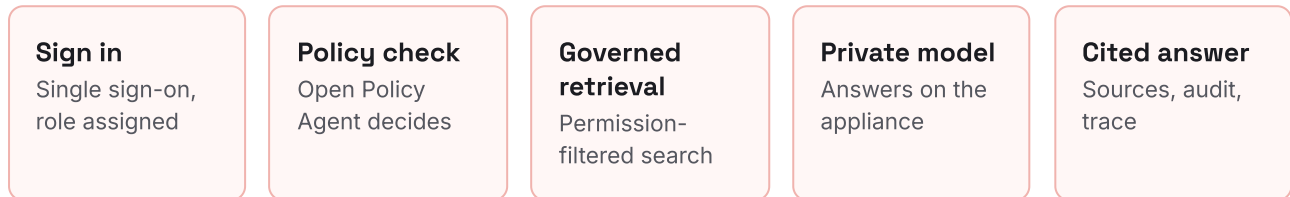


Illustration · The Security Center: policies, access, activity and audit in one place.

From request to result, inside your walls.

Open WebUI for the chat experience, open-weight language models served on the appliance, Open Policy Agent for policy decisions, OIDC single sign-on, and self-hosted Langfuse for observability, all wrapped in a Logic Sonata governed gateway and Security Center.



01 **Connect and classify**

Documents are ingested through a secure conversion pipeline and tagged with owner, folder, classification and lifecycle.

02 **Index privately**

A hybrid keyword and vector index is built on the appliance. Nothing is sent to an external service.

03 **Ask in a familiar chat**

Users sign in with single sign-on and ask questions in a clean chat interface, in their own language.

04 **Check every request**

The policy engine evaluates the user, role and content before retrieval, and again before results are shown.

05 **Answer with sources**

The model answers only from permitted passages and cites each source.

06 **Audit and observe**

Every request, denial and model call is traceable in the Security Center.

What the Private Knowledge Assistant does.

Answers with citations

Every answer links back to the source document and passage, so users can verify it in one click.

Permission-filtered retrieval

Access rules are applied before retrieval and results are re-authorised before display. Restricted content fails closed.

Single sign-on and roles

Federates with your identity provider over OIDC. Roles such as user, manager, auditor and administrator, plus approved temporary access.

Central policy engine

Policies are managed in one place, enforced by Open Policy Agent, and can be simulated before they go live.

Security Center

One console for policies, access requests, security events, AI activity, traces, model usage, audit timeline and reports.

Content defence

Detects prompt-injection attempts and poisoned documents before they can influence an answer.

Multilingual hybrid search

Keyword and semantic retrieval combined, qualified for multilingual content across regional teams.

Controlled egress

Outbound traffic passes an approved egress proxy. Prompts, documents and answers stay on the appliance.

Appliance operations

Encrypted data at rest, signed offline updates with rollback, backup, restore and health monitoring.

Controls built in, not bolted on.

Every Logic Sonata system runs on infrastructure you control. Data boundaries, access and audit are designed in from the start rather than added later.

- ✓ Role-based access with fail-closed denial for restricted folders
- ✓ Result re-authorisation and cache isolation between users
- ✓ Prompt-injection and poisoned-content defence
- ✓ Signed model catalogue: only approved models can run
- ✓ Approved egress proxy and hardened containers
- ✓ Encrypted storage, signed updates and tested recovery

Data boundary

Prompts, documents, images and results are processed on your appliance or dedicated private infrastructure. They are not sent to public AI services and are never used to train third-party models.

Inside the apparel demo.

Our live demonstration runs a fictional apparel manufacturer with production, quality, technical design, sales, HR, finance and executive teams. The same question gets a different, correct answer depending on who asks.

Standard user

Gets cited answers from production, quality, technical and sales documents and the staff handbook. Payroll, budgets and board strategy are denied and the attempt is logged.

Manager

Sees everything a standard user sees. Restricted HR, finance and executive content stays denied unless access is explicitly delegated.

Administrator

Governs policies, access requests and temporary access in the Security Center, with every change recorded on the audit timeline.

Auditor

Reviews AI activity, traces, denials and reports without being able to change policy.

Where it pays off first.

Production and quality SOPs

Inspection levels, defect standards and line procedures answered on the factory floor.

Technical design and tech packs

Construction details, measurement specs and material requirements in seconds.

Buyer requirements and compliance

Customer manuals, testing protocols and compliance rules, with the exact clause cited.

HR handbook and onboarding

Staff get policy answers any time; payroll and personal data stay restricted.

Sales and customer service

Product knowledge and order procedures for faster, consistent responses.

Management content, locked down

Finance, payroll and board material available only to authorised roles, and every attempt logged.

Measuring the pilot

We agree success measures before the pilot starts. Typical measures include:

- ✓ Time to answer common SOP and policy questions
- ✓ Share of answers with verified citations
- ✓ Reduction in repeat questions to supervisors and HR
- ✓ Policy denials and access requests handled through the Security Center

Deploy where your data needs to live.

On-premise

Dedicated hardware inside your building for the most sensitive data. Nothing leaves your network.

Hosted private

Dedicated capacity on managed private GPU infrastructure, isolated from public AI services. No hardware purchase.

Hybrid

Sensitive workloads on-site, everything else hosted, under one set of policies and one audit trail.

Hardware, sized to the workload

Logic Sonata is hardware-neutral. Smaller teams often start with a compact AI workstation such as NVIDIA DGX Spark or an AMD Ryzen AI Max+ system, both with up to 128 GB of unified memory in a desktop-sized unit. Larger rollouts use multi-GPU servers or a private cloud cluster.

A proven rollout path

Assess

Review current AI use, data exposure and priority use cases. Deliver a 30/60/90-day roadmap.

Pilot

Deploy one working use case with real users and agreed success measures.

Roll out

Extend to more teams with governance, integration and role-based training.

Manage

Monitoring, model updates, knowledge refreshes and quarterly reviews.

One partner is accountable for the whole stack: hardware, software and models, your company knowledge, access controls, implementation, training and ongoing support.

Questions to ask any vendor.

Use this checklist when comparing options, including ours.

- ☐ Are permissions enforced before retrieval, or only filtered in the final answer?
- ☐ Does every answer cite its source, and can users open the passage in one click?
- ☐ What happens when a user asks about restricted content? Is the attempt recorded?
- ☐ Can policies be simulated before they go live?
- ☐ Does the system federate with our existing identity provider?
- ☐ How are prompt injection and poisoned documents detected?
- ☐ Where do prompts, documents and logs physically reside?
- ☐ How are updates signed, tested and rolled back?

Frequently asked questions

Can users see documents they are not allowed to?

No. Permissions are checked before retrieval and again before results are displayed. If a user asks about restricted content, the request is denied and recorded.

Does it work with our existing sign-in?

Yes. The assistant federates with your identity provider over OIDC, so staff use the accounts they already have.

Which document types are supported?

Common office formats such as PDF, Word, Excel, PowerPoint and text are converted through a secure ingestion pipeline. We confirm your specific formats during the assessment.

Private AI, built inside your walls.

Logic Sonata delivers secure private AI for businesses in Singapore, Vietnam, Indonesia, Malaysia and Thailand. Each solution combines hardware, software, company knowledge, access controls, implementation, training and ongoing support, deployed on-premise, hosted or hybrid.

Our product portfolio: Private Knowledge Assistant, Private Vision Intelligence, Private Coding Assistant, Private AI Agent Platform and Private Image Generation Studio.

Next step: a 30-minute consultation

We will show the Private Knowledge Assistant working, discuss your data and requirements, and scope a pilot around your use case.

www.logicsonata.com/contact

sales@logicsonata.com

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